

TrES-3b

R-0496

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180831 · created 2026-07-20T12:51:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458361.788 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.033
Transit depth (Rp/Rs)^2	3.79 +/- 1.28 [%]
Orbital Inclination (inc)	81.68 +/- 1.28
Airmass coefficient 1 (a1)	1.004 +/- 0.021
Airmass coefficient 2 (a2)	-0.0018 +/- 0.0055
Scatter in the residuals of the lightcurve fit is	2.13 %
Best Comparison Star	#3 - [572, 393]
Optimal Aperture	3.44
Optimal Annulus	8.53
Transit Duration (day)	0.055 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.195 $\pm$ 0.033 vs 0.16309 (deviation 0.69 σ)
Duration fit vs published	0.055 d vs 0.0567 d (deviation 0.08 σ)
Mid-time O–C	-1.8 min
Depth SNR	4.2
Residual scatter	2.13 %

Light curve

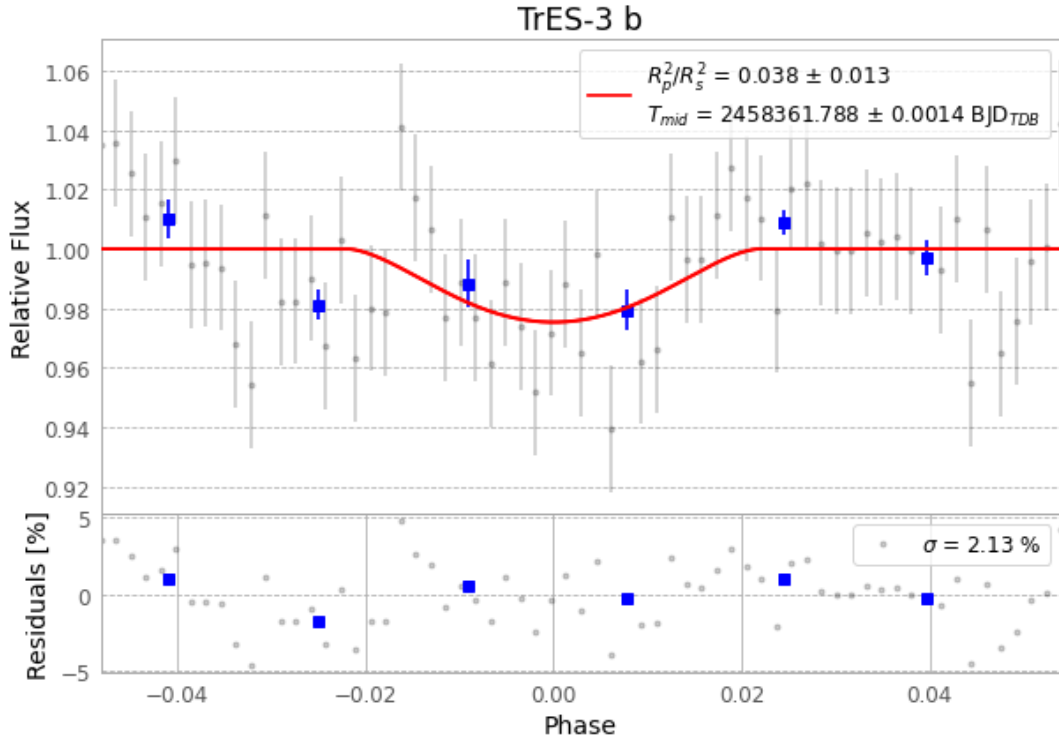


Figure 1 — FinalLightCurve_TrES-3 b_2018-08-30.png (SHA-256 b6799cf1efb68205...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [248, 310], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 60cf6d396ff81b8e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

65 FITS frames (43 MB), TRES-3180831052112.FITS → TRES-3180831083312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180831.log (2f6f1994b72b...), FinalLightCurve_TrES-3 b_2018-08-30.png (b6799cf1efb6...), FinalLightCurve_TrES-3 b_2018-08-30.csv (9c822fa9d93e...)

Compute resources

Wall time 110.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 12:51Z.